



**Cape Peninsula
University of Technology**
creating futures

FACULTY OF INFORMATICS & DESIGN

Advanced Diploma in Information & Communication Technology

PROJECT 4

SUBJECT GUIDE:	2020
COURSE CODE:	PRJ470S / PRJ471S / PRJ472S
NQF LEVEL:	7
PREREQUISITE:	PRT360S / PRT361S / PRT362S or Equivalent

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ORGANISATIONAL COMPONENT

1. INTRODUCTION

Project 4 is one of the compulsory subjects that must be completed in the one-year Advanced Diploma course. This offering is intended to provide you with an opportunity to integrate learning across the one-year curriculum and deliver an integrated project that reflects the appropriate academic standard for the level (NQF 7). Aspects of the project will be delivered through individual effort while other aspects will be achieved through teamwork and collaboration. It is the responsibility of the learner to ensure the successful implementation of the project. This is a group project with a maximum of six students in a group. Students must form the group, choose a team leader and email it to the lecturer. It is suggested that you name the team. The team lead will communicate with the lecturer on behalf of the team. **It is strongly suggested that you form an interdisciplinary team (team consisting of App. Dev, MM and Comm. Net.) so they can put their corresponding domain knowledge towards the final Artefact.**

The purpose of this Learner Guide is to provide you with written guidelines as to what to expect of the University and what the University in turn expects of you, with regard to this course.

2. GENERAL

2.1 Contact Information

DESIGNATION	NAME	LOCATION	TELEPHONE NUMBER	E-MAIL ADDRESS
Subject Co-ordinator	Amlan Mukherjee	Engineering 2.27A	(021) 460-3700	mukherjeea@cput.ac.za
Lecturer	Amlan Mukherjee	Engineering 2.27A	(021) 460-3700	mukherjeea@cput.ac.za **
Domain Coordinator	Will be available with Ver1.2 of the subject guide			
Head of Department	B. Kabaso	Engineering 2.30	(021) 460-3713	kabasob@cput.ac.za
Administrative Assistant	N Allie	Secretary IT	(021) 460-3010	allieN@cput.ac.za
Support Technician	R. Pietersen	Engineering 1.4	(021) 460-4209	pietersenr@cput.ac.za
Support Technician	I. Saliem	Engineering 1.28	(021) 460-3031	saliemi@cput.ac.za
Support Technician	C. Sindayigaya	Engineering 1.28	(021) 460-3031	sindayigayac@cput.ac.za

** PLEASE NOTE ANY EMAIL TO THE LECTURER MUST START WITH THE SUBJECT LINE "PROJECT 4 2020:..."

e.g. "PROJECT 4 2020:Request to join team meeting on atin"

2.2 Class Times and Venues

The classes are mainly based on consultations rather than theory classes. Students are expected to have team meetings and keep the minutes for the meetings. The lecturer can join team meetings on requests via email.

2.3 Learner Management System (LMS)

<https://myclassroom.cput.ac.za/>: Blackboard used for general communications and announcements. It is the responsibility of the learner to check Blackboard regularly.

3. STUDY MATERIALS

3.1 Prescribed Textbook

No prescribed book for this subject. The books from other subjects can be used to gather necessary knowledge.

Teaching Methods:

The lecturer will discuss during the first 2 lectures on how to kick off a project in class. Later most of the examples and notes will be posted on the Black Board.

Group discussions with individual groups are more welcome during consultation hours. The lecturer will prefer to meet the entire group during discussion.

Portfolio of Evidence and Logbook: A portfolio of evidence showing all the work done during the year including the final project need to be produced. A log book to show day to day activities may also be required.

3.2 Recommended Readings

Lecturer will post links / notes throughout the year via Black Board.

4. MODULE CREDITS

The subject carries 20 credits and delivered at NQF 7 level. (please note: It will be confirmed in V1.2 of the subject guide)

Please refer to the Faculty prospectus on

<http://www.cput.ac.za/academic/faculties/informaticsdesign/prospectus> for further information.

4.1 Planning of Time Allocation for Learning Activities

There are 200 notional hours allocated to this module. (Also will be confirmed via V1.2 of the subject guide)

5. ETHICAL CONDUCT

5.1 Roles and Responsibilities

Each student in the group must contribute equally towards the final deliverable. He / She will follow the rubrics assign tasks to individual students in order to complete the work. A form will be made available via Black Board. The team leader will complete the form. All members must sign the form and the team leader should return the form to the lecturer.

Participation is an important part of the group meetings. Participation means both attendance and engagement.

Grievance procedures: A learner's lecturer is the first person that should be informed, if there is a problem/dissatisfaction with something in the class. Thereafter, the domain leader and as a last resort, the head of Department.

Academic dishonesty will NOT be tolerated in any form whatsoever. This includes submitting someone else's work as your own (plagiarism) and copying someone else's program. Learners caught cheating will be given a form to fill in and thereafter face the disciplinary committee. A committee will then discuss the situation and make recommendations to the head of Department. Where two or more learners' handed-in programs seem undeniably to be identical copies of each other, ALL the learners involved will receive a mark of ZERO and an additional punishment will be given after the hearing with the disciplinary committee.

Learners with special needs: CPUT is keen to accommodate learners with special needs. These needs are catered for without influencing academic standards. It is, however, the learner's responsibility to inform the lecturer in good time of any special needs in order to ensure effective communication in this regard.

Learners are requested to make appointments to meet with the lecturer during the available consultation times. As this will not always be possible, however, lecturer is also expected to make time to meet with learners in 'emergency' situations.

Kindly refer to CPUT's Policy Docs and Student Guidelines:

https://www.cput.ac.za/storage/study_at_cput/Student_Rules%20Regulations_2018.pdf

5.2 Class attendance and participation

Kindly refer to page 18 of CPUT's General Handbook Academic and Student Rules and Regulations:

https://www.cput.ac.za/storage/study_at_cput/Student_Rules%20Regulations_2018.pdf

5.3 Grievance procedures

Kindly refer to page 61 of CPUT's General Handbook Academic and Student Rules and Regulations:

https://www.cput.ac.za/storage/study_at_cput/Student_Rules%20Regulations_2018.pdf

5.4 Plagiarism

Kindly refer to page 30 of CPUT's General Handbook Academic and Student Rules and Regulations:

https://www.cput.ac.za/storage/study_at_cput/Student_Rules%20Regulations_2018.pdf

**NB: An amended version of the General Handbook will be emailed to you during the first term.*

6. ASSESSMENT

6.1 Assessment policy and regulations

Assessment will be taken through Black Board uploads, presentations and running of the Artefact. The lecturer will communicate the date and time for submission of work on Black Board and schedule the presentation during the project week of June and October exams.

ASSESSMENT RULES

The nature of your assessments may consist of one of, or a combination of the following:

- Pdf/ Presentation / Running of the Artefact
- Oral / Practical presentations

The lecturer will communicate which nature of assessment will be used for each assessment.

Assessment briefs:

- The brief will outline all the specifications and requirements for a particular assessment and it will also include an outline of the criteria by which your learning will be judged. The type of assessment you will be doing in each of the terms will be discussed with you in detail during a classroom session. The brief will be handed out timeously before the assessment due date.
- Copies of all briefs will be posted on Blackboard.
- Because the Information Technology industry is deadline driven, it is very important that you learn to adhere to deadlines, thus all assessment due dates in this subject are **NON-NEGOTIABLE**. The programme has specific penalties for the late submission of any task. It is very important that you become familiar with those rules and consult your lecturer well ahead of time if there are any reasons why you might not be able to meet your assessment deadline.
- **Penalties for the late submission of tasks will be strictly enforced.**
- Group assignments/work forms a large part of this subject because, amongst others, it aims to encourage and develop teamwork, group dynamics, participation, effective time management, conflict management and creativity.
- Your ability to demonstrate professional conduct during class and in your interaction with the lecturer/s will also be observed and in some cases assessed.
- **ALL TEAM MEMBERS MUST BE PRESENT DURING PRESENTATIONS.**

6.2 Assessment Opportunities: Administration

Please check the rubrics in Appendix A, B, C and D. The lecturer can change it as per the requirement of the project and will communicate to the student. Please note that 80% of the marks will be based on

individual contribution and 20% will be based on group work. Depending on the project scope it may vary as 70% for individual work and 30% for group work.

There will be one assessment per Term.

Assessment Type	Percentage	Moderated	Assessment
Assessment 1 Submission of PDF in BB	10% (T1)	No	Project Initiation Rubric which includes scope of the project, risk analysis, SMART analysis and the work breakdown structure(WBS) Due Date: End of day 27th March 2020 Refer to Appendix A
Assessment 2 Individual presentation within the group which must adhere with the WBS submitted on or before 27 th March, 2020. Submission of the Presentation Via Black Board, Detailed design and test plans	30% (T3)	Yes	Detailed design Presentation. Test Plan Creation: Refer to Appendix B Presentation: During Project week Upload: On or before midnight 31st May 2020
Assessment 3 Submission of individuals work, Proof of execution of Test Cases	20% (T2)	No	Build and upload the unit you are working on as per the WBS. Proof of execution of the Test plans. Refer to Appendix C Upload: On or before 28th Aug 2020
Assessment 4 Presentation of the integrated Artefact via group presentation and running of the Artefact. Evaluate the results, communicate your findings from the evaluation in form of an Article (In IEEE format) Upload the Artefact and the Article via Black Board. If the Artefact is big then submit it via a Flash or CD.	10% (Group Presentation) 30% (Individual contribution in achieving the goal) (Can be 20% and 20% depending on the project)	Yes	Do a presentation (PPT). Group introduction and brief of the project by the team lead for 2 min. Then 5 min presentation per team member followed by running of the Artefact for 5 min. Then 10 min for answering questions from the lecturer / board. The questioners will be both to the group and individual. Refer to Appendix D Upload: On or before 23rd Oct 2020 Presentation: During project week

STUDY COMPONENT

7. SUBJECT SPECIFICATIONS

7.1 Purpose of the Subject

This subject is intended to provide learners with an opportunity to integrate learning across the second year curriculum and deliver an integrated project that reflects the appropriate academic standard for the level. The course presents an understandable, integrated view of the many concepts skills, tools, and techniques involved in completing

7.2 Subject Structure

TERM	TOPIC AND CONTENT DISCUSSION	SPECIFIC OUTCOMES
1	Project Initiation	Refer to rubric in Appendix A
2	Project Planning and Design	Refer to rubric in Appendix B
3	Project Execution	Refer to rubric in Appendix C
4	Project Closeout	Refer to rubric in Appendix D

7.2.1 Subject Outcomes

SUBJECT OUTCOMES	DESCRIPTION
SO1	How to work in an interdisciplinary team and contribute towards achieving the final goal of the project.
SO2	How to create a design goal of a project
SO3	To understand the design criteria for a project
SO4	How to build and test an Artefact
SO5	To understand redesigning and testing

7.2.2 Subject Breakdown Schedule

The lecturer will decide and communicate with the student.

7.3 Articulation with other subjects in the programme

Project 4 is the practical component for the implementation of the theory in the Advanced Diploma course of study and furthermore across the programme. They will produce an Artefact at the end of the year which will be added to their portfolio of evidence.

7.4 Learning presumed to be in place

- A degree in IT *Competent in written and verbal communication*
- Reading and writing skills.
- Computer literate

7.5 Critical Cross-field Outcomes

CRITICAL CROSS-FIELD OUTCOMES	DESCRIPTION
CCFO1	Identify and solve problems with responses which display responsible decisions, using critical thinking, have been made.
CCFO2	Work effectively with others as a member of a team, group, organisation or community.
CCFO3	Organise and manage oneself and one's activities responsibly and effectively.
CCFO4	Collect, analyse, organise and critically evaluate information.
CCFO5	Communicate effectively, using visual, mathematical, and/or language skills in the modes of written and/or oral presentation.
CCFO6	Use science and technology effectively and critically, showing responsibility towards the environment and the health of others.
CCFO7	Demonstrate an understanding of the world as a set of related systems by recognising that problem-solving contexts do not exist in isolation.
CCFO8	Contribute to the full personal development of each learner and the social and economic development of the society at large, by making it the underlying intention of any programme of learning to create awareness in individuals about the importance of: i) reflecting on and exploring a variety of strategies to learn more effectively; ii) participating as responsible citizens in the life of local, national and global communities; and iii) being culturally and aesthetically sensitive across a range of social contexts. (RSASAQA, 2001:24)

8. STUDY UNITS/STUDY THEMES

The lecturer will decide and communicate with the student.

8.2 Assessment Opportunities and Assessment strategy

The lecturer will decide viewing the scope of the project and communicate with the student.

8.3 Subject Outcomes and Associated Assessment Criteria

Subject Objectives and Scope of the Subject (Outcomes)
On successful completion of this subject, you will be able to: 1. Define the scope of a project. 2. The deliverables of the project are described in terms of the expected scope. 3. The project is planned according to the selective processes, resources and timing criteria to comply with the project life cycle. 4. Work effectively in an interdisciplinary team. 5. Co-ordinate the processes over various life cycle phases. 6. Communicate the result on completion of the project

Appendix A:**Rubric Advanced Diploma****Weight 10% (T1)****Due Date: End of day 27th March 2020.**

PRT470 / 471 / 472S		2020											
Project Initiation Rubric												Weight %	
The Team are rated according to each of the following variables (using a scale from 1 to 5).													
		Unsatisfactory	Satisfactory	Good	Excellent	Total	Score		T110				
Marking Scheme		0	1..2	3..4	5								
Task	1. Criteria: What is the Project all About?												
1.1	Why are you doing this work? Describe the desired end result of this project.					5							
1.2	What specific outcomes will be achieved, and how will you measure these outcomes?					5							
1.3	Scope: What are the boundaries for this project					5							
1.4	Deliverables: What will the project deliver as outputs? include a date that each deliverable is expected. You'll use this information to monitor milestones. (Individual members must specify what they will deliver in T2.)					5							
1.5	Constraints: What things must you take into consideration that will influence your deliverables and schedule?					5							
1.6	Assumptions: What assumptions are you making at the start of the project?					5							
TOTAL						30							
Task	2. Risk Analysis												
2.1	Risk Identification: Identify the risks within the project, and that you'll either need to manage or accept.					5							
2.2	Risk Prevention: Describe what you are going to do to mitigate or manage risks.					5							
2.3	Risk Management: Where you can't prevent risks, what are your contingency plans for dealing with them? What actions will you take should the risk materialize?					5							
TOTAL						15							

Task	3. SMART ANALYSIS		
3.1	Specific:		
	3.1.1 Is the objective precise and well-defined?	5	
	3.1.2 Will its meaning be clear to anyone who reads it?	5	
3.2	Measureable:		
	3.2.1 How will we know when the objective has been achieved?	5	
	3.2.2 What evidence will be needed to confirm this? How will you judge this evidence?	5	
3.3	Attainable:		
	3.3.1 Can this be done at all? If so, is it within your capabilities?	5	
	3.3.2 Are sufficient resources available to achieve it?	5	
3.4	Realistic:		
	3.4.1 How sensible is it in the current economic and technological environment?	5	
	3.4.2Is it possible for the individuals who will work on the objective to accomplish it?	5	
	3.4.3 Does it fit into our overall strategy?	5	
3.5	Time-Limited:		
	Can we do it within the given timeframe?	5	
TOTAL		50	
Task	4. The Work Breakdown Structure (WBS)		
Continue to break the work down until a task list is developed which meets the following criteria,(one and only one owner can be assigned to each of the lowest level tasks)			
4.1	Clearly defined outputs are evident for each task	5	
4.2	Quality can be monitored through performance criteria associated with each output	5	
4.3	The tasks communicate the work to be accomplished to the person who is accountable	5	
TOTAL		15	
TOTAL FOR TERM		110	

T1**10**

Appendix B: Will be available in Ver1.2 of the study guide

Appendix C: Will be available in Ver1.2 of the study guide

Appendix D: Will be available in Ver1.2 of the study guide

STUDENT ACKNOWLEDGEMENT

CAPE PENINSULA UNIVERSITY OF TECHNOLOGY
STUDENT ACKNOWLEDGEMENT
2020
PRT470S/PRT471S/PRT472S

ACKNOWLEDGE THAT YOU HAVE READ THE SUBJECT GUIDE AND UNDERSTAND ITS CONTENT

Please read the following statement and complete the spaces that follow. Make a copy of this page for your own reference. Hand-in this original page during the next class session.

I have read the subject guide of **Project 4** and understand the information and my responsibilities as a learner for this subject. I am also aware that the Subject Guide may be revised due to unforeseen circumstances and it remains my responsibility to adhere to the most recent guide.

Name & Surname:

(Print)

Student Number:

Contact number: (home/res) Contact number: (cell)

Signature:

Date: